# Pythagorean Theorem: Which Side Goes Where?
## Answer Key

Grade 8 / Pre-Algebra

**Quick Answers**

| | | | |
|---|---|---|---|
| 1. 10 | 3. 15 | 5. 41 | 7. 56, -56 |
| 2. 12 | 4. 13 | 6. 10 | 8. 40 |

**Curriculum**

**Level:** Grade 8 / Pre-Algebra
8.G.B.7, 8.G.B.8 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
*Difficulty 1–5 of 5 across 8 tagged checks.*

**Common wrong answers**

1. *If they got 14: added the legs instead of adding their squares*
2. *If they got 13.93: substituted the hypotenuse 13 into a leg slot and added the squares*
3. *If they got 18.79: put the ladder length 17 in a leg slot and added the squares, so the wall came out longer than the ladder*
4. *If they got 10.91: subtracted the squares of the two coordinate differences, as if the segment were a leg*
5. *If they got 38.97: treated the longer leg 40 as the hypotenuse and subtracted the squares*
6. *If they got 5.29: called the larger coordinate difference the hypotenuse and subtracted the squares*
8. *If they got 87.08: treated the 80 m side as the hypotenuse and subtracted the squares to get the diagonal*

**The habit this sheet builds:** name the hypotenuse *before* substituting. $a^2 + b^2 = c^2$ is only true with $c$ across from the right angle, so a known hypotenuse means subtract, and a computed leg must always land shorter than the hypotenuse. Every wrong answer below comes from the same slip: a hypotenuse dropped into a leg slot, or a leg promoted to hypotenuse.

### 1. Legs 6 cm and 8 cm — find the hypotenuse.

Both givens are legs, so the unknown is the hypotenuse $c$ and the theorem is used in its adding direction.

$$c^2 = 6^2 + 8^2 = 36 + 64 = 100 \qquad c = \sqrt{100}$$

The answer is longer than either leg, as a hypotenuse must be: $\boxed{10\text{ cm}}$.
*Watch for:* $6 + 8 = 14$. The theorem adds the *squares*, never the sides.

## 2. Hypotenuse 13 cm, one leg 5 cm — find the other leg.

Here the hypotenuse is given, so it belongs alone on the $c$ side of the equation and the missing side is a leg. Write the theorem, then isolate the unknown leg $b$:

$$5^2 + b^2 = 13^2 \quad \Rightarrow \quad b^2 = 13^2 - 5^2 = 169 - 25 = 144$$

So $b = \sqrt{144}$, giving $\boxed{12 \text{ cm}}$, which is shorter than the 13 cm hypotenuse — the check a leg must pass.
*Watch for:* $\sqrt{13^2 + 5^2} \approx 13.93$, which puts the hypotenuse in a leg slot and returns a leg longer than the hypotenuse.

## 3. Find the mistake: the 17-foot ladder.

(a) The ladder *is* the hypotenuse: it runs from the ground to the wall, opposite the right angle where wall meets ground. So the height it reaches must be less than 17 feet. Jae's 18.79 feet is longer than the ladder itself, which is impossible before any arithmetic is checked.
(b) The mistake is substitution: Jae put the hypotenuse 17 into a leg slot and added. With the hypotenuse known, subtract instead:

$$h^2 = 17^2 - 8^2 = 289 - 64 = 225 \qquad h = \sqrt{225}$$

The ladder reaches $\boxed{15 \text{ ft}}$, safely shorter than the 17-foot ladder.

## 4. Distance from $A(2,3)$ to $B(7,15)$.

Drop a horizontal and a vertical segment from $A$ and $B$ to build a right triangle. Those two are the legs; $\overline{AB}$ is the hypotenuse, because it lies opposite the right angle where they meet.

$$\text{run} = 7 - 2 = 5 \qquad \text{rise} = 15 - 3 = 12 \qquad AB^2 = 5^2 + 12^2 = 25 + 144 = 169$$

So $AB = \sqrt{169} = \boxed{13 \text{ units}}$, longer than either leg.
*Watch for:* $\sqrt{12^2 - 5^2} \approx 10.91$ — subtracting treats $\overline{AB}$ as a leg, but the segment joining the two points is always the hypotenuse.

## 5. Find the mistake: legs 9 cm and 40 cm.

(a) The problem says both 9 and 40 are *legs*. The hypotenuse is not one of the givens — it is the side being asked for. "Longest given" does not mean "hypotenuse"; only position relative to the right angle decides that. Sam's answer is also shorter than a side he was told is a leg, which is impossible.
(b) Two legs given means add:

$$c^2 = 9^2 + 40^2 = 81 + 1600 = 1681 \qquad c = \sqrt{1681}$$

The hypotenuse is $\boxed{41 \text{ cm}}$, just longer than the 40 cm leg, as expected.

**6. Find the mistake: distance from $P(-3, 2)$ to $Q(5, 8)$.**
(a) Kim's two differences are the *legs* of the right triangle — one horizontal, one vertical. Neither can be the hypotenuse, because they meet at the right angle. The hypotenuse is $\overline{PQ}$ itself, the side she was asked to find, so the squares must be added, not subtracted. Her 5.29 is even shorter than both legs, which no hypotenuse can be.
(b) Compute the legs from the coordinates, then add the squares:

$$\text{run} = 5 - (-3) = 8 \qquad \text{rise} = 8 - 2 = 6 \qquad PQ^2 = 8^2 + 6^2 = 64 + 36 = 100$$

So $PQ = \sqrt{100} = \boxed{10 \text{ units}}$.

**7. The 65-inch television.**
(a) The diagonal is the hypotenuse of the right triangle formed by the screen's height and width, so the equation given is the theorem with the hypotenuse correctly isolated. Solve it:

$$x^2 + 33^2 = 65^2 \quad \Rightarrow \quad x^2 = 4225 - 1089 = 3136$$

Taking square roots gives two solutions, because squaring erases a sign: $\boxed{x = 56 \text{ or } x = -56}$.
(b) A width is a length, and lengths are never negative, so $x = -56$ is discarded and the screen is 56 inches wide. Note the sanity check: 56 is less than the 65-inch diagonal, as a leg must be.

**8. How much shorter is the diagonal walk?**
Rosa's route follows two sides: $80 + 60 = 140$ meters. Tam's route is the diagonal of the rectangle, which is the hypotenuse of a right triangle whose legs are the two sides:

$$d^2 = 80^2 + 60^2 = 6400 + 3600 = 10000 \qquad d = \sqrt{10000} = 100$$

The diagonal is 100 meters, and the saving is the difference between the two routes: $140 - 100 = 40$.
Tam's walk is $\boxed{40 \text{ m}}$ shorter.
*Watch for:* treating the 80 m side as the hypotenuse and subtracting, which gives a "diagonal" of about 52.92 m — shorter than a side of the rectangle it crosses — and a saving of about 87.08 m, more than either side.